6

Formalism

Easing us from informal ideas into formal mathematics. This chapter will motivate that move, develop more formal approaches to the structures we've already seen, acclimatize us to the formalism, and see what we get out of it.

6.1 Types of tourism

It is possible to visit a country whose language you don't speak, and still have a culturally rewarding trip, in which you expand your mind and don't just behave like an obnoxious tourist demanding that everyone speak your language to you. Perhaps it is less accepted that it is possible to "visit" the world of math in this way, without having any formal understanding and without knowing any of the technical language, but still appreciating the culture and being enriched by it. This sort of visit to category theory is more or less the idea of my book *How to Bake* π; perhaps I would push the analogy a little further and say that that book also contained some basic phrases as an introduction to the language.

In the present book, however, we're aiming to take things a little further. I want to provide a way in to the true language, a route that is approachable but is still a decent beginning that doesn't just consist of memorizing how to say "Good morning", "Thank you" and "My name is Eugenia".

I have already been gradually ramping up the level of formalism, by gradually introducing technical mathematical notation, much of which is actually category theory notation. But I want to take a little pause and talk directly about formalism and notation because I know that this is one of the offputting aspects of learning math. Mathematicians get very fluent and take it for granted, which makes them forget to explain it sometimes. It's a bit like how it can be very hard to get a native speaker of a language to explain the grammar, if they know it too instinctively.

If you're already comfortable with formal mathematical notation you can probably skip or skim this chapter. However, the formal mathematical notation you've seen might not be used in quite the same way as in abstract mathematics, so I'd suggest reading this chapter quickly anyway.

6.2 Why we express things formally

One of the important issues around formalism is why we do it. If you don't find something inherently interesting and you also don't see the point of it then it will almost certainly be offputting. If you find it fun then it doesn't really matter whether or not you think it has a point, but if you find it offputting then understanding why it's helpful is at least a starting point.

The formalism of mathematics is like a language specifically designed for what we are trying to do. Normal language is geared towards expressing things about our daily human experience, and less towards making rigorous arguments about abstract concepts. So our normal language falls short when we're trying to do math.

In fact, even among normal languages, different ones express different things in different ways. People who speak several languages often find themselves missing words and expressions in one language when they're speaking another. In English I miss the concise and pithy four-character idioms that my family often uses in Cantonese; one of my favorites, which somehow doesn't translate quite so pithily into multi-syllabic English, is something like "rubbish words covering the sky" to indicate that someone is spouting utter nonsense.

Formal mathematical language and notation are partly to do with efficiency, and partly to do with abstraction (and those two concepts are themselves related). The efficiency is to do with when we keep talking about the same thing over and over again, in which case it's really more efficient to have a quick way to refer to it. If you're only going to mention it once then it doesn't matter so much. If you come to visit Chicago for a week you might talk about driving up the highway that runs alongside the Lake. However, if you live here it might get tedious to keep saying "the highway that runs alongside the Lake" so it helps to know it's called Lakeshore Drive, and then if you really talk about it a lot you might just start calling it "Lakeshore" or "the Drive" or (somewhat amusingly) LSD.

Mathematical language and notation happens similarly. If you eat two cookies and then another three cookies and then never do anything similar again in your life then you might never need any more language or notation. But if you ever plan to develop thoughts about that sort of thing any further then it can help to write the numbers 2 and 3 rather than the words "two" and "three". It's quicker, takes up less space, and is faster for our brains to process. We can then also move from writing "2 and 3" to $2 + 3$ for similar reasons. I would then argue that it makes it easier to see further analogies such as the analogy between $2 + 3$ and 2×3.

This is where the part about abstraction comes in. Concise notation can help us to package up multiple concepts into a single unit that our brains can treat as a new object, which is one of the crucial aspects of abstraction.

For example the expression on the right is fine and precise and rigorous, but it is long-winded.

$$2 + 2 + 2 + 2 + 2$$

So we make the following notation and get the idea that multiplication is repeated addition.

$$5 \times 2$$

Note that this is not the *only* way to think of multiplication. Some people think this means we should never talk about multiplication as repeated addition, but I think it's more productive to see the sense in which it *can* be thought of in this way, while also being open to how we can generalize if we don't *insist* on it being thought of in this way.

Anyway, we have now made new notation for *iteration* of an operation, and this created a new operation. We could now repeat that process, that is, we could apply that entire thought process to the new operation $\times$, treating it as analogous to the original operation $+$. This might not have been so obvious if we hadn't used this formal notation.

If we iterate multiplication we get things like this:

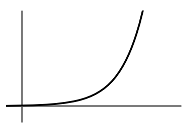
$$5 \times 5 \times 5$$

Here is the notation that mathematicians created for that, making a new operation called exponentiation.

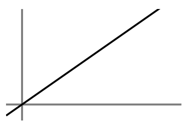
$$5^3$$

At time of writing, we have just all gone into lockdown over the COVID-19 virus. Viruses spread by exponentiation, because each infected person infects a certain number of people on average. Thus the number of infected people is multiplied by that factor repeatedly, producing exponentiation. Without that precise language and notation it would be much harder to reason with the concept and understand it.

As it is, mathematicians understand that exponentiation makes numbers grow extremely rapidly, and it's not just that they grow rapidly, but the growth itself gets faster and faster as it goes along, as shown here.



This is very different from repeated addition, where things might grow fast but they never get any faster, as shown by the straight line graph.



This understanding is why scientists and mathematicians know that advance precautions are critical when exponential growth is involved, even if things seem to be moving very slowly at that particular point in time. The formalism of math helps us to model the exponential in question even when it's early on in the slow part of the growth, and predict what will happen in the future if we don't take steps to slow it down. To people who don't understand exponentials this has unfortunately seemed like panic and over-reaction.

When we're doing research and we come up with a new concept, one of the first things we do is give it a name and some notation, so that we can move the concept around in our heads more easily. It's a bit like when a baby is born and one of the first things we do is give it a name.

I'm now going to take a few of the examples from the previous chapters and introduce the formality that mathematicians would use to discuss them.

6.3 Example: metric spaces

In Section 4.1 we thought about a type of distance measured "as the taxi drives" (in a grid system, like in many American cities) as opposed to "as the crow flies". Different types of distance in math are called *metrics*. This more abstract technical-sounding word may sound alien but it serves the purpose of

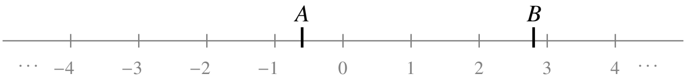
a) reminding us that we're not thinking of straightforward distance, and
b) opening our minds to the possibility of generalizations which have some things in common with distance but aren't actually distance.

Here is how mathematicians write that down formally. We start by thinking about what "distance" really means, right down to its bare bones. At the most basic, it is some process where you take two locations, and produce a number.

In order to think about this more carefully we're going to use some "place-holders" to refer to those things. This is a bit like in normal life when we're talking about two random people, and we might call them person A and person B. This is only necessary if things get complicated, say we're discussing an interaction and we need to keep clear about who's doing what. So we might say "one person makes the coffee and the other person gets the cookies and gives them to the person who made the coffee and the person who made the coffee takes the coffee round while the person who makes the cookies cleans up". That is long-winded, and we might sum it up more clearly as "A makes the coffee, B gets the cookies and gives them to A, then A takes the coffee round while B cleans up". We could make it even more efficient with a table:

A	B
makes coffee	gets cookies and gives to A
takes coffee round	cleans up

Now for our case of distance we're talking about *locations* A and B. If we were just going to talk about features of them individually that would be enough, but we want to talk about the distance between them, which is a number involving both of them. It's a bit like a subtraction, but not quite. If A and B were just numbers on a number line



we could find the distance between them just by doing a subtraction $B - A$ (or $A - B$ if A is bigger). But if they're positions in 2-dimensional space or higher then it's a bit more complicated. Anyway we want to separate out the notation for a type of distance from the instructions for how to calculate it. $B - A$ is more of an instruction or a method of calculating the distance between them. So we could say this:

$$\boxed{\text{distance between } A \text{ and } B \text{ on number line} = B - A}$$

Now we get to the efficiency/abstraction part. If we write "distance between A and B on number line" a lot, we might get bored of doing it. Also it's a lot of stuff for our eyes to take in.

So we might shorten it to this, for example: $d(A, B) = B - A$

Here the d stands for distance and the parentheses don't indicate any kind of multiplication; they are just a sort of box in which to place the two things whose distance apart we're taking.

Technicalities

We'll later see that basically what we're doing here is defining a *function* and that this notation using parentheses is a standard way to write down what happens when you apply a function to an input. In this case the function happens to have two inputs, A and B. Other functions like $\sin$ just take one input and then the output is written as $\sin(x)$.

We can now *generalize* this by a sort of leap of faith: even if we have yet to work out how we're going to calculate a distance, we can still use the notation $d(A, B)$ for the distance between A and B. In formal mathematical notation we make a declaration saying that's what we're going to do. It's a bit like saying "I hereby declare that I am going to use the notation $d(A, B)$ to mean the taxi-cab distance between A and B". This is still a bit imprecise because we haven't said what A and B are allowed to represent. Are they numbers or are they people or are they locations? So the full declaration might look like this:

Let A and B be locations on a grid. We will write $d(A, B)$ for the taxi-cab distance between A and B according to the grid.

Now we might want to get even more abstract and talk about new types of distance that we haven't even defined yet, and then we want to say in what circumstances something deserves to count as a type of "distance".

That's like what happens if some random person comes along and says "Hey, I've discovered a treatment for COVID-19" — we'd want (I hope) to do some tests on it to see if it really deserves to be called a treatment, and the thing we'd have to do before doing those tests is decide what the criteria are for being counted as a treatment.

So if some random person comes along and says "Hey, I've discovered a new type of distance", mathematicians want to have some criteria for deciding if it's going to count or not. Here are some criteria.

Criterion 1: distance is a number

The first criterion is that distance should be a way of taking two locations and producing a number. So given any locations A and B, $d(A, B)$ is a number. There are different kinds of number though, so if we're being really precise we should say what kind of number. We probably want distances to go on a continuum, so we don't just want whole numbers or fractions, but *real numbers*: the real numbers include the rational numbers (fractions) and irrational numbers, and sit on a line with no gaps in it. So, for any locations A and B, $d(A, B)$ is a real number.

Criterion 2: distance isn't negative

We typically think of distance as having size but not direction — it's just a measure of the gap between two places. This means that we shouldn't get negative numbers. So in fact for any locations A and B, $d(A, B) \geq 0$.

Criterion 3: zero

Can the distance between places ever be 0? Well yes, the distance from A to A should be 0. In our formal notation: $d(A, A) = 0$. But is there any other way that distance can be 0? Usually not, but in fact there are more unusual forms of distance where it might be possible. We don't usually include those in *basic* notions of distance though. So now what we're saying is "the distance from A to B can only be 0 if $A = B$".

Note on A and B being the same

It might be confusing to have locations A and B that turn out to be the same one. Let's go back for a second to the example of the people A and B who were sorting out the cookies and the coffee. Usually in normal life if we talk about person A and person B we really are talking about two entirely different people. But if the group needing cookies and coffee was short-staffed that day then it's possible one person could take on both roles, in which case person A and person B would both be the same person.

In math even if we use two different letters for things there is always the possibility that they could represent the same thing. For example if you write x for the number of sisters you have and you write y for the number of brothers you have it's quite possible that those are the same number. Writing those variables as x and y means that the numbers *could be* different but it still allows for the possibility of them being the same. Whereas if you write them both as x then you have declared in advance that they're the same and ruled out the possibility of them being different.

If we really want to rule out the possibility of them being the same in math we call them "distinct", so instead of saying "Let A and B be locations" we would have to say "Let A and B be distinct locations". That is equivalent to adding in an extra condition saying "$A \neq B$" and mathematicians don't like extra conditions if we can avoid them, so it's usually preferable or more satisfying to see if we can include the possibility of A and B being equal.

In summary: for distance we have decided that the distance from a location to itself is 0, and that this is the only way we can have a 0 distance. In formal language we say: $\boxed{d(A,B) = 0 \text{ if and only if } A = B.}$

I will discuss more about what "if and only if" means in the next section.

Criterion 4: symmetry

We said in Criterion 2 that we were just going to measure the gap between places, without direction being involved. We have specified that this means the answer is always a non-negative number, but that's not enough: we need to make sure that the "distance from A to B" doesn't come out being different from the "distance from B to A".

That is, for any locations A and B we have this: $\boxed{d(A,B) = d(B,A).}$

This is called *symmetry* because we have flipped the roles of A and B. Later we'll look at symmetry more and more abstractly. Symmetry starts off as being about folding things in half and the two sides matching up, but eventually we generalize it to being about doing a transformation on a situation and it still looking the same afterwards. That's what symmetry is about here.

Criterion 5: detours

The last criterion is about taking detours. This is the one that you might not immediately think up if you're just sitting around thinking about what should count as a sensible notion of distance, but here it is: if you stop off somewhere along the way it should not make your distance any *less*. At best it won't change the distance, but it might well make the journey longer. Now let's try putting that into formal mathematics.

It might help to draw a diagram: we're trying to go from A to B but we go via some other place X like in this triangle.

$$\begin{tikzcd} & & X \arrow[dr, dashed] & \\ A \arrow[urr, dashed] \arrow[rrr] & & & B \end{tikzcd}$$

The direct journey has this distance:

$$d(A, B)$$

whereas the detour has this distance:

$$d(A, X) + d(X, B)$$

Now the detour distance should be "no less" than the direct distance:

$$d(A, X) + d(X, B) \geq d(A, B)$$

Note that "no less" is logically the same as "greater than or equal to", so in the formal version we use the sign $\geq$.

This condition is often referred to as the *triangle inequality* because it relates to the triangle picture above, and produces a specific inequality relationship. In math, when we refer to things as an inequality we usually don't just mean that two things are unequal; after all, most things are unequal. Usually we are specifically referring to a relationship with $\geq$ or $>$ in. The point is that even if we can't be sure two things are equal, sometimes it helps to know for sure that one is bigger, or definitely not bigger. See, math isn't just about equations: sometimes it's about inequations.

That's the end of the formal criteria for being a type of distance in math. It is quite typical that we now give it a formal name, both to emphasize that we've made a formal definition, and to remind ourselves that some things that aren't physical distance can count as this more abstract type of distance. In this case, we call it a *metric* in math. We have carefully put some conditions on something to say when it deserves to be called a metric. Crow distance and taxi distance are both examples. Sometimes the money or energy spent getting from one place to another could be examples too.

Once we have the definition with a set of criteria, aside from working with it rigorously and finding more examples, we can think about generalizing it by relaxing some criteria. We might think there are some interesting examples that *almost* satisfy all the criteria but not quite, and maybe they deserve to be studied too. Rather than just throw them out or neglect them, mathematicians prefer to make a slightly more relaxed notion that will include them, and then study that. It often makes the theory more difficult because things aren't so rigidly defined, but it often also makes it more interesting, and anyway, it's more inclusive.

Things To Think About

T 6.1 Can you think of some measures that are a bit like distance as we defined it above but that fail some of these criteria? See if you can think of some that fail each criterion in turn, except maybe the first one.

6.4 Basic logic

While we are talking about the formalism of mathematics it's just as well to make sure we're clear about the basic logic underpinning it. As we are going to be building arguments using rigorous logic, the way in which basic logic flows is going to be critically important, although a deep background in formal logic is definitely not needed. Really what matters is following the logic of the arguments I make as we go along, so even if you don't feel you understand all of this section out of context, you could keep going and just come back to it if you want more clarification about a later argument.

Logic is based on *logical implication*. Given any statements A and B we can make a new statement "A implies B" which means "if A is true then B has to be true", or, to put it more succinctly "if A then B". The fact that it's a *logical* implication means that the truth of B follows by sheer logic, not by evidence, threats or causation. This usually means it is sort of inherent from some definitions.

We sometimes write "implies" as a double arrow like this: $\boxed{A \Longrightarrow B.}$ This means we have fewer words to look at, and also helps emphasize the directionality of logic. One source of confusion is that in normal language we use the following types of phrases interchangeably (in vertical pairs):

if A is true then B is true	if you are a human then you are a mammal
B is true if A is true	you are a mammal if you are a human

This looks like we've changed the order around, but we haven't changed the direction of the *logic*, we've just changed the order in which we *said* it: the statements are logically equivalent. Note that some statements are *logically* equivalent without being *emotionally* equivalent, sort of like the fact that Obamacare and The Affordable Care Act are the same thing by definition, but induce very different emotions in people who hate Obama so much that they immediately viscerally object to anything that has his name in it. This means that in normal life we need to choose our words carefully.

In math we are interested in things that are logically equivalent, but we still choose our words carefully because sometimes there are different ways of stating the same thing that make it easier or harder to understand, or shed different light on it that can help us make progress.

In summary we have these several ways of stating the same implication, and they sort of go in pairs (horizontally here) corresponding to ways of stating the thing in the opposite order without changing the direction of the logic:

A implies B	B is implied by A
if A then B	B if A
$A \Longrightarrow B$	$B \Longleftarrow A$

I particularly like using the arrows here because to my brain it's the clearest way of seeing that the left- and right-hand versions are the same, without really having to think. Or rather, I don't have to use logical thinking, I can just use a geometrical intuition.

We are going to use a lot of notation involving arrows as that is one of the characteristic notational features of category theory.

The arrow notation also makes it geometrically clear (at least to me) that if we change the way the arrow is pointing at the actual letters then something different is going on, as in this pair.

$$\begin{aligned} A &\Longrightarrow B \\ B &\Longrightarrow A \end{aligned}$$

This reversing of the flow of logic produces the *converse* of the original statement. The converse is logically independent of the original statement. This means that the two statements might both be true, or both false, or one could be true and the other false. Knowing one of them does not help us know the other. Here are some examples of all those cases.

Both $A \Longrightarrow B$ and $B \Longrightarrow A$ are true

A = you are a legal permanent resident of the US
B = you are a green card holder in the US

Both $A \Longrightarrow B$ and $B \Longrightarrow A$ are false

A = you are an immigrant in the US
B = you are undocumented[†]

One is true but the other is false

A = you are a citizen of the US
B = you can legally work in the US

As a statement and its converse are logically independent we have to be care-

[†] You can be born in the US but not have any documents about it.

ful about not confusing them. But if they are both true then we say they are logically equivalent (even if they aren't emotionally equivalent) and say

A is true *if and only if* B is true.

Mathematicians like abbreviating "if and only if" to "iff" in writing sometimes, but this is hard to distinguish from "if" in speaking out loud so it should still be pronounced "if and only if" out loud. The phrase "if and only if" encapsulates both the implication and its converse:

A if B	means	$A \Leftarrow B$
A only if B	means	$A \Rightarrow B$

The last version has a tendency to cause confusion so if it confuses you it might be worth pondering it for a while. My thought process goes something like this: A implies B means that whenever A is true B has to be true, which means that A can only be true if B is true.

We have this full table of equivalent ways of saying the same statement, including a double-headed arrow for "if and only if":

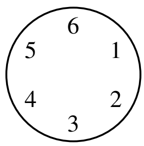
original		converse	
$A \Rightarrow B$	$B \Leftarrow A$	$B \Rightarrow A$	$A \Leftarrow B$
A implies B	B is implied by A	B implies A	A is implied by B
if A then B	B if A	if B then A	A if B
A only if B	only if B then A	B only if A	only if A then B

logical equivalence
$A \Leftrightarrow B$
A if and only if B
A iff B

I think that the notation with the arrows is by far the least confusing one and so I prefer to use it.

6.5 Example: modular arithmetic

In Section 4.2 we looked at some worlds of numbers in which arithmetic works a bit differently. One of the examples was the "6-hour clock".

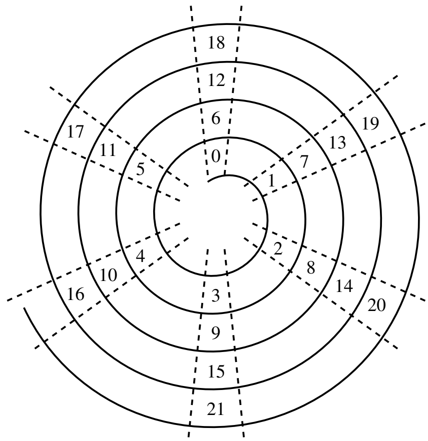


We started making a list of some of the numbers that are "the same" as each other on this clock, such as these.

7	is the same as	1
8	is the same as	2
9	is the same as	3

Now the thing is that we can't just sit and make a list of *all* possible numbers that are the same as each other, because there are infinitely many. In situations where it is impossible to write everything down one by one it can be extremely helpful to have a formal and/or abstract way of writing things, so that we can give a sort of recipe or make a machine for working every case out, without actually having to write out every case one by one.

We then drew this spiral diagram showing what happens if you keep counting round and round the clock and writing in the numbers that live on the "same hour" as each other. If you pick one position maybe you can spot a relationship between all the numbers in that position. In the 0 position it's perhaps the most obvious — all the numbers in that position are divisible by 6.



If you look at the 1 position perhaps you can see that the numbers jump by 6: $1 \longrightarrow 7 \longrightarrow 13 \longrightarrow 19 \longrightarrow \cdots$. This jumping by 6 happens in each position on the clock. This makes sense as we're "wrapping" an ordinary number line around a circle — each time it goes round it wraps six numbers around, so we will get back to the same place six later. What we've informally found is this: *Two numbers live in the same position provided they differ by a multiple of* 6.

Now let's try and express that in formal mathematics. The first step is to refer to our two numbers as something, say, a and b. As with the locations in the previous section, we leave open the possibility that a and b are actually the same number. Of course, if they're actually the same number then they obviously are going to live in the same position on the clock, and hopefully this will still make sense when we've put our conditions on.

Next we want to see if they "differ by a multiple of 6", which is a condition on their difference $b-a$. Like with distance we're really thinking about the gap and ignoring whether it's positive or negative, so we could also say $a - b$ but for various (slightly subtle) reasons mathematicians prefer using $b - a$.

Now how are we going to say "is a multiple of 6"? Well, we don't know what multiple it's going to be, so we could use another random letter, say k, to represent a whole number, and then a multiple of 6 is anything of the form $6k$. However this extra letter is a bit annoying, so mathematicians have invented this notation instead: $6 \mid m$ means "6 divides m", or "6 goes into m", which is the same as saying m is a multiple of 6, just the other way round.

So we have the following more formal characterization:

$$\boxed{a \text{ and } b \text{ live in the same position provided } 6 \mid b - a.}$$

We have been referring to this as a and b "living in the same position" on the 6-hour clock" but in mathematics this is called "congruent modulo 6" or "congruent mod 6" for short, and we write it like this: $a \equiv b \pmod 6$.

The idea is that we're using notation that is reminiscent of an equals sign, because this is in fact a bit like an equation, it's just an equation in a different world — the world of the 6-hour clock. This is technically called the "integers modulo 6" and is written $\mathbb{Z}_6$. We can now try writing out an entire definition of how congruence works in $\mathbb{Z}_6$. One more piece of notation will help us: we write "$a \in \mathbb{Z}$" as a shorthand for "a is an element of $\mathbb{Z}$", that is, a is an integer. Then we have this:

Definition 6.1 Let $a, b \in \mathbb{Z}$. Then $a \equiv b \pmod 6$ if and only if $6 \mid b - a$.

You might notice that I've used the words "if and only if" rather than the symbol $\Longleftrightarrow$ here, despite me having said that I like the clarity of the symbols. The difference is a subtle point: here we're making a definition of the thing on the left via the thing on the right, rather than declaring two pre-existing things to be logically equivalent. This feels slightly different and is less symmetric as a statement, and under those circumstances I prefer the words to the symbol. These are often the sorts of subtleties that mathematicians don't say out loud but just *do* and hope that people will pick them up.

One of the things that I think becomes clearer by this formulation (although it took us a while to get here) is how to generalize this to other clocks, that is, to the integers modulo n for other values of n; all we have to do is replace every instance of 6 with n. Here n can be any positive integer, and I might write the positive integers† as $\mathbb{Z}_+$.

Definition 6.2 Let $n \in \mathbb{Z}_+$ and $a, b \in \mathbb{Z}$. Then $a \equiv b \pmod n$ iff $n \mid b - a$.

We'll now do a small proof to see what it looks like when we use this formalism. It's good to try and do it yourself first even if you don't succeed.

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† Some people refer to the positive integers as the natural numbers, written $\mathbb{N}$, but there is disagreement about whether the natural numbers include 0 or not. While some people have a very strong opinion about this I think it's futile to insist on one way or the other as everyone disagrees. I think there are valid reasons for both ways and so the important thing is to be clear which one you're using at any given moment. Some people like worrying about things like this; I'd rather save my worry for social inequality, climate change and COVID-19.

Things To Think About

T 6.2 Can you show that if $a \equiv b$ and $b \equiv c$ then $a \equiv c \pmod{n}$?

To work out how to prove something it is often helpful to think about the intuitive idea first, before turning that idea into something rigorous that we can write down formally. The idea here is that if the gap between a and b is a multiple of n and the gap between b and c is also a multiple of n, then the gap between a and c is essentially the sum of those two gaps, so is still a multiple of n. Nuances of what happens when c is actually in between are taken care of by the formal definition of "gap". This is the informal idea behind this formal proof. Proofs are not always in point form but I think this one is clearer like that. Note that the box at the end means "the end".

Proof We assume $a \equiv b$ and $b \equiv c$, and aim to show that $a \equiv c \pmod{n}$.

- Let $a \equiv b \pmod{n}$, so $n \mid b - a$.
- Let $b \equiv c \pmod{n}$, so $n \mid c - b$.
- Now if $n \mid x$ and $n \mid y$ then $n \mid x + y$.
- Thus it follows that $n \mid (c - b) + (b - a)$, which is $c - a$, so $n \mid c - a$.
- That is $a \equiv c$ as required. □

Things To Think About

T 6.3 On our n-hour clock we just drew the numbers 0 to $n - 1$. How do we know that every integer is congruent to exactly one of these?

Here is the formal statement of that idea. (A lemma is a small result that we prove, smaller than a proposition, which is in turn smaller than a theorem.)

Lemma 6.3 *Given $a \in \mathbb{Z}$, there exists b with $0 \leq b < n$ and $a \equiv b \pmod{n}$.*

The idea is that we are "wrapping" the integers round and round the clock. Every multiple of n lands on 0 and then the next $n - 1$ integers are a "left-over" part, which wraps around the numbers 1 to $n - 1$. This is behind the idea of "division with remainder", where you try and divide an integer by n, and then if it doesn't go exactly you just say what is left over at the end instead of making a fraction. This sounds complicated but is really what children do *before* they hear about fractions, and it's what we do any time we're dealing with indivisible objects like chairs: you put the leftovers to one side.

I think this is an example where the idea is very clear but the technicality of writing down the proof is a little tedious and unilluminating, so I'm not going to do it. But it's worthwhile to try it for yourself to have a go at working with formality; at the same time it's not a big deal if you don't feel like it.

Often when we get comfortable with abstract ideas we work at the level of ideas rather than the level of details. This sounds unrigorous but we can put in technical work early on to make that level rigorous. This is what category theory is often about, which is one of the reasons I love it so much: I feel like I'm spending much more time working at the level of ideas, which is my favorite level, rather than the level of technicalities.

This is a slightly hazy way in which mathematicians working at a higher level might seem unrigorous to those who are not so advanced, as they are more comfortable taking large steps in a proof knowing that they *could* fill them in rigorously. It's like a tall person being able to deal with stepping stones that are much further apart.

6.6 Example: lattices of factors

In Chapter 5 we drew some diagrams of lattices of factors of a particular number, using arrow notation for the factor relationships. The arrow notation is a lot like the vertical bar notation we used in the previous section, it's just a bit more general, a bit more flexible, and a bit more geometric. In the previous section we wrote a divides b as $a \mid b$ as that's how it's typically done if we're not thinking about category theory. Writing it $a \longrightarrow b$ instead is only superficially visually different from the vertical bar version, but that visual difference opens the door to drawing geometric diagrams, which would have been impossible or impractical and confusing using the vertical bar.

Different formal presentations are possible even when we're using the same kind of abstraction, and in this sense although formality and abstraction are related, they're not quite the same issue. We are going to see that category theory does both of these things in a productive way:

- it comes up with a really enlightening form of abstraction using the idea of abstract relationships, and also
- it comes up with a really enlightening formal presentation of that abstraction, using diagrams of arrows.

To warm up into the first part of this we're going to spend the next chapter investigating a particular type of relationship that will lead us into the definition of category in the chapter after that. From this point on we are going to start using the more formal notation and language that we've discussed in this chapter, so you might find yourself needing to go more slowly if you're unfamiliar with that kind of formality. Mathematics really does depend on that formality and so I think we need it in order to get further into category theory than just skimming the surface.